WORKED SOLUTION

STEP III 2001, Question 7

Pure • Orthogonal Trajectories

STEP III 2001, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q7

The key idea

The differential equation is separable, so solving it shows directly what the family of curves is. For the orthogonal curve, perpendicular tangents mean the gradients multiply to -1. That gives a second separable equation, and when we integrate it the function from the opening sketch turns up. The first sketch is there so that we can draw the final curve almost for free.

Solution

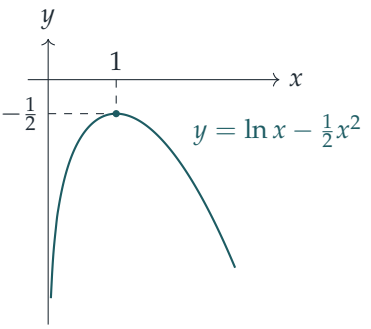
The graph of $\ln x - \frac{1}{2}x^2$

Let $f(x) = \ln x - \frac{1}{2}x^2$, defined for $x > 0$. Then

$$f'(x) = \frac{1}{x} - x = \frac{1 - x^2}{x}, \qquad f''(x) = -\frac{1}{x^2} - 1 < 0.$$

For $x > 0$ the only stationary point is at $x = 1$, where $f(1) = -\frac{1}{2}$. Since $f'' < 0$ everywhere, this is a maximum and the graph is concave throughout. So $f(x) \leq -\frac{1}{2} < 0$ for every $x > 0$, and the graph never meets the x-axis.

At the ends of the domain, $f(x) \to -\infty$ as $x \to 0^+$ (because $\ln x \to -\infty$), so the y-axis is a vertical asymptote. Also $f(x) \to -\infty$ as $x \to \infty$, because $\frac{1}{2}x^2$ grows much faster than $\ln x$.



The family of parabolas

The constant function $y = 0$ is a solution. Elsewhere, we separate the variables, working away from $x = \pm 1$, where the equation is undefined:

$$\int \frac{\mathrm{d}y}{y} = \int \frac{2x}{x^2 - 1}\,\mathrm{d}x \quad \Longrightarrow \quad \ln|y| = \ln|x^2 - 1| + c.$$

So $|y| = \mathrm{e}^c\,|x^2 - 1|$. Absorbing the sign into the constant, and including $y = 0$, the solutions are

$$y = A(x^2 - 1), \qquad A \text{ constant.}$$

Conversely, each such curve satisfies the equation, since

$$y' = 2Ax = \frac{2x \cdot A(x^2 - 1)}{x^2 - 1} = \frac{2xy}{x^2 - 1}.$$

For $A \neq 0$ each curve is a parabola. It passes through $(1, 0)$ and $(-1, 0)$ because $x^2 - 1 = 0$ there. Its gradient $2Ax$ is zero only at $x = 0$, so its vertex is $(0, -A)$, which lies on the y-axis. This is the required family.

The orthogonal curve

Take a point (x, y) with $x \neq \pm 1$ and $y \neq 0$. Exactly one parabola of the family passes through it, namely the one with $A = y/(x^2 - 1)$, and its gradient there is $\dfrac{2xy}{x^2 - 1}$. For the required curve to cross that parabola at right angles, the product of the two gradients must be -1, so the curve must satisfy

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x^2 - 1}{2xy} = \frac{1 - x^2}{2xy}.$$

Ask yourself Why is it legitimate to use the parabolas' differential equation, rather than the formula $y = A(x^2 - 1)$, to find the gradient?

The differential equation gives the gradient of the family member through (x, y) in terms of x and y alone, with no reference to A. That is exactly what we need, because the orthogonal curve meets a different parabola at each of its points.

Separating the variables again,

$$2y\,\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x} - x \quad\Longrightarrow\quad y^2 = \ln x - \tfrac{1}{2}x^2 + C,$$

taking $x > 0$ because the curve passes through $(1,1)$. The right-hand side is $f(x) + C$, the function we sketched at the start. At $(1,1)$ we need $1 = 0 - \frac{1}{2} + C$, so $C = \frac{3}{2}$ and the curve is

$$\boxed{y^2 = \ln x - \tfrac{1}{2}x^2 + \tfrac{3}{2}} \qquad \text{or equivalently} \qquad y^2 = f(x) + \tfrac{3}{2}.$$

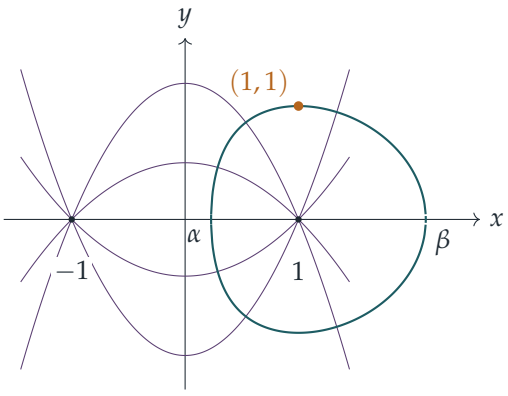
Strictly, a solution of the differential equation through $(1,1)$ is the upper half $y = \sqrt{f(x) + \frac{3}{2}}$. Its reflection in the x-axis also cuts every parabola orthogonally, so the natural answer is the whole closed curve.

Sketching the curve

Real points need $f(x) \geq -\frac{3}{2}$. Our first sketch shows that f rises from $-\infty$ to its maximum of $-\frac{1}{2}$ at $x = 1$ and then falls back to $-\infty$. So $f(x) = -\frac{3}{2}$ has exactly one root $\alpha \in (0,1)$ and one root $\beta > 1$. Numerically, $\alpha \approx 0.229$ and $\beta \approx 2.123$. Therefore:

- the curve lies in the strip $\alpha \leq x \leq \beta$ and is symmetric about the x-axis;
- y^2 is largest, equal to 1, at $x = 1$, giving the points $(1, \pm 1)$, where the tangent is horizontal;
- at $(\alpha, 0)$ and $(\beta, 0)$ we have $y^2 = 0$ while $\mathrm{d}(y^2)/\mathrm{d}x = f'(x) \neq 0$, so $\mathrm{d}y/\mathrm{d}x$ is infinite and the tangents are vertical.

So the curve is a closed oval that is lopsided: it rises steeply on the left and slopes off more gently on the right. Some of the parabolas are drawn in for comparison.



If we allowed $x < 0$ and replaced $\ln x$ by $\ln|x|$, the same working would also produce the mirror-image oval in the left half-plane. That oval does not pass through $(1,1)$, so it is not part of the answer.

Checks

- At $(1,1)$ the curve's gradient is $\frac{1-1}{2} = 0$. No parabola passes through $(1,1)$ itself, since every parabola meets the line $x = 1$ only at $(1,0)$. Close to $x = 1$ the parabolas are nearly vertical, which fits with a horizontal tangent on the curve.
- Numerically, at $x = 0.5$ the curve has $y = \sqrt{\ln 0.5 - 0.125 + 1.5} \approx 0.826$ and gradient $\frac{0.75}{2(0.5)(0.826)} \approx 0.908$. The parabola through that point has gradient $\frac{2(0.5)(0.826)}{-0.75} \approx -1.101$, and $0.908 \times (-1.101) \approx -1$.
- We confirmed orthogonality at several thousand random points of the oval, and computed the roots $\alpha \approx 0.2291$ and $\beta \approx 2.1226$ by bisection.

Common slips

- Using the gradient of one fixed parabola, such as $2Ax$ with a particular A, instead of the gradient of the family member through the general point (x, y).
- Forgetting that $y^2 = f(x) + \frac{3}{2}$ needs $f(x) \geq -\frac{3}{2}$, and so drawing an unbounded curve rather than a closed oval.

Beyond STEP

Finding a family of curves that meets a given family at right angles is the problem of *orthogonal trajectories*. In physics these appear as field lines and equipotentials, for example in electrostatics or two-dimensional fluid flow. The real and imaginary parts of an analytic complex function always give two such families, which is one route into conformal mapping. Boyce and DiPrima, *Elementary Differential Equations*, covers separable equations and orthogonal trajectories. Needham, *Visual Complex Analysis*, shows the complex-variable picture.

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